

CYBERSECURITY AND DIGITAL CITIZENSHIP

1. Cybersecurity

Cybersecurity refers to the practice of protecting computer systems, networks, devices, and digital information from theft, damage, unauthorized access, misuse, and cyberattacks.

Objectives of Cybersecurity

- Protect personal and organizational data.
- Prevent unauthorized access.
- Reduce the risk of malware and cyberattacks.
- Maintain the safe and reliable use of digital systems.

2. Concept of Cybercrime

Cybercrime refers to criminal activities carried out using computers, digital devices, communication networks, or the Internet.

Examples of Cybercrime

- Hacking and unauthorized access
- Phishing and online fraud
- Identity theft
- Cyberbullying
- Malware attacks
- Data theft
- Online financial scams

Nepal and cybercrime: The Electronic Transactions Act, 2063 is an important Nepalese legal framework related to electronic transactions and cyber-related offences. It provides legal provisions for addressing certain offences committed through electronic means.

3. Prevention Methods from Cybercrime

3.1 Use Strong Passwords

A strong password is a difficult-to-guess password that uses a suitable combination of letters, numbers, and symbols and is preferably long.

Tips for Using Strong Passwords

- Avoid simple sequences such as 123456 or abcdef.

- Include uppercase and lowercase letters.
- Include numbers and special symbols where allowed.
- Avoid common dictionary words and easily guessed information.
- Use longer passwords or passphrases.
- Do not reuse the same password for important accounts.

3.2 Multi-Factor Authentication (MFA)

MFA is a security method that requires two or more different types of evidence to verify a user's identity before granting access.

Factor	Meaning	Examples
Something you know	Information known by the user	Password, PIN
Something you have	A physical or digital item possessed by the user	Smartphone, security badge, authenticator device
Something you are	A biometric characteristic of the user	Fingerprint, voice, eye/retina recognition

3.3 Software Updates

A software update installs improvements, bug fixes, and security patches. Regular updates help protect devices from known security weaknesses.

3.4 Authentication

Authentication is the process of verifying that a user is really the person or account they claim to be.

- Passwords and PINs
- One-time passwords (OTP)
- Security keys or verification codes
- Biometric authentication

3.5 Use of Firewalls

A firewall is a hardware or software security system that monitors and controls incoming and outgoing network traffic according to defined security rules.

3.6 Safe Web Browsing Techniques

- Use a secure network connection, especially when handling sensitive information.
- Keep the web browser updated and use a reputable browser.
- Use trusted search engines and carefully evaluate search results.
- Prefer secure websites that use HTTPS when entering sensitive information.
- Avoid suspicious links, downloads, pop-ups, and unknown websites.

4. Digital Citizenship

A digital citizen is a person who uses the Internet, digital devices, and online services responsibly, safely, ethically, and respectfully.

Guidelines for Good Digital Citizenship

- Think before posting or sharing anything online.
- Avoid sharing sensitive personal information publicly.
- Use privacy and security settings appropriately.
- Use different or unique passwords for important accounts and change compromised passwords promptly.
- Report unlawful, harmful, abusive, or inappropriate online behaviour.
- Check information before forwarding or reposting it.

5. Concept of Netiquette

Netiquette: Netiquette is derived from “network” and “etiquette.” It refers to the rules and guidelines for polite, respectful, and appropriate behaviour while communicating online.

Internet ethics: Internet ethics focuses on the responsible and morally appropriate use of online resources, information, communication, and digital communities.

Examples of Good Netiquette and Online Behaviour

- Respectful communication — communicate politely and avoid unnecessary insults.
- Avoid offensive content — do not post hateful, abusive, or harmful material.
- Respect privacy — do not expose another person's private information without permission.
- Do not share personal information carelessly.
- Avoid spamming or flooding — do not send repeated or unwanted messages.
- Give credit — acknowledge the creator when sharing another person's work.
- Think before posting — consider the possible consequences of a post.
- Do not repost abusive behaviour — avoid spreading harmful content even when criticizing it.
- Accept constructive criticism — respond calmly to useful feedback.
- Maintain a positive online presence — behave responsibly and respectfully.

6. Digital Footprint and Privacy

A digital footprint is the record of information created by a person's online activities, such as searches, posts, comments, uploads, registrations, and website visits. It is also called a digital shadow or electronic footprint.

Types of Digital Footprint

Type	Definition	Example
Active digital footprint	Information deliberately created or shared by the user.	Posting on social media, writing comments, uploading photos, filling in online forms.
Passive digital footprint	Information collected automatically or indirectly as a person uses digital services.	Browsing history, IP-related records, cookies, device or usage information collected by websites.

6.1 Online Privacy

Online privacy is the ability and practice of controlling how personal information is collected, used, stored, and shared while using digital services and the Internet.

- Use privacy settings carefully.
- Limit unnecessary personal information shared online.
- Use strong passwords and MFA.
- Be cautious with location sharing.
- Review app and website permissions.
- Think about the long-term effect of online posts.

Advantages of Digital Footprint

- Health monitoring — digital records can support tracking of health and activity information.
- Social connection — online activity can help people maintain relationships and communities.
- Targeted advertising — digital activity can help services present relevant products and content.
- Law enforcement — digital records can sometimes support investigations when legally obtained and appropriately used.

Disadvantages of Digital Footprint

- Security vulnerabilities — stored information can become a target for attackers.
- Legal and ethical concerns — data collection and use may raise privacy, consent, and ownership issues.
- Exploitative business practices — personal data may be used in ways users do not fully understand or expect.
- Cybercriminal exploitation — exposed information can be used for fraud, impersonation, or other attacks.
- Spying and online abuse — excessive tracking or misuse of personal information can harm users.

Practical Steps to Protect Yourself from Digital-Footprint Risks

1. Know security rules — understand the security and privacy practices of the services you use.
2. Update privacy settings — limit who can see your information and posts.
3. Secure your devices — use screen locks, passwords, MFA, and appropriate security software.
4. Remove old accounts — close unused accounts and delete unnecessary personal information where possible.
5. Update software — keep operating systems, browsers, and applications patched.
6. Disable unnecessary location tracking — turn off location access for apps that do not need it.